CS-548: Cloud-native Software Architectures

Computer Science Department • University of Crete
Instructor: Antony Chazapis

Website: https://www.csd.uoc.gr/~hy548/

Version: 202601

Assignment 3: Autoscaler, Helm

1. (*40% grade*) An example of a complete "hello world" service is available at GitHub (https://github.com/chazapis/hy548/blob/master/examples/hello-ingress.yaml). Extend the YAML:
    a. To limit each Pod to a maximum of 20% CPU and 256MB RAM.
    b. With a HorizontalPodAutoscaler manifest, which will increase the number of Pods in the Deployment when the average CPU usage exceeds 80%. Set a minimum of 1 Pod and a maximum of 8 for the Deployment.

    Then, run some http benchmark to find the maximum requests per second the service can handle, using 1 or 100 simultaneous clients. At how many containers does scaling stop? Provide the new YAML, the commands used to test in minikube, the test results, and a screenshot of the tool you used, or its output if it was a command line utility.

2. (*40% grade*) Following on from the previous exercise, create a Helm chart for your modified "hello world" service. The chart should define variables for:
    a. The string to return (use the `MESSAGE` environment variable).
    b. The endpoint to use for the service (currently `/hello`).
    c. The CPU and memory limits of each Pod (optional, default is no limits).
    d. The maximum number of Deployment Pods for the HorizontalPodAutoscaler (optional, default is 10).

    Provide the chart files and the commands needed to install a service named "goodbye" that will show the message "Goodbye world!", use the `/goodbye` endpoint, be limited to 25% CPU (no memory limit), and scale automatically up to 20 Pods via the HorizontalPodAutoscaler.

3. (*20% grade*) If you have enabled the ingress addon in minikube, remove it. Issue the commands to install the Ingress controller implemented with Nginx using Helm. You will find the chart at https://artifacthub.io/packages/helm/ingress-nginx/ingress-nginx. Try again deploying one of the services of the above exercises. What changes are needed in the YAML files to make them work (if any)?

Notes:

- The assignment is personal.
- All exercises contribute equally to the overall grade (unless individual percentages are defined).
- Ask for questions and clarifications at the beginning of lectures or via email.
- Write down your answers in a Markdown-formatted text file in either Greek or English.

- Place all files in a folder created for the assignment. Submit your answers in the private GitHub repository that you will use for the course. Share the repository with the instructor (username "chazapis").
- Commits done after the exercise's deadline will be ignored.